

BOOKNOOK BOOKS

End-to-End Data Engineering, Analysis & Visualization

Project Report and Reproducibility Guide

Capstone Project • PostgreSQL • Python • Pandas • NumPy • Matplotlib • Seaborn • Folium • PyWaffle

Project Members:

- Muhammad Ali Waris Khan
- Umar Nadeem
- Muhammad Ali Raja
- Muhammad Mutaqqi Khalid
- Muhammad Ahsan
- Muhammad Huzaifa

1. Project Overview

BookNook Books is an end-to-end data project demonstrating how raw bookstore data can be transformed into a structured PostgreSQL database and then analyzed and visualized with Python. The workflow is: create the database structure, clean and transform data, load it into PostgreSQL, query the database, calculate business metrics, and communicate findings through visualizations. The project follows ETL—Extract, Transform, Load—using CSV inputs, pandas transformations and PostgreSQL storage. The analysis layer uses SQL and pandas, NumPy supplies numerical calculations, and the visualization libraries communicate the results.

2. Project Objectives

- Build four related tables: customers, products, orders and orderitems.
- Clean and transform customer, product, order and order-item data.
- Preserve relationships through the appropriate IDs.
- Query live PostgreSQL data with pandas.
- Calculate line_total as quantity × unit price.
- Use NumPy for mean, standard deviation and above-average count.
- Create trend, comparison, distribution and relationship charts.
- Use Seaborn, PyWaffle and Folium for Day 3 storytelling.

3. Overall Workflow

1. Database foundation — create/reset the PostgreSQL tables.
2. Customer/product ETL — clean and load master data.
3. Order ETL — validate customer/date fields and load valid orders.
4. Order-item loading — load orderitems after valid order IDs exist.
5. Day 2 analysis — SQL JOIN, pandas transformation, NumPy statistics and charts.
6. Day 3 storytelling — Seaborn, Waffle, Folium and written interpretations.

4. Database Design and ETL

The four-table relational design is the foundation. Orders connect customers to purchases, while orderitems connect orders to products. This structure supports clean JOINS and avoids unnecessary duplication.

The customer/product stage prepares stable master data. Consistent identifiers are important because later order matching depends on them. Orders require validation because an order must match a valid customer and have a usable order date. A reproducibility warning is important: the order-item seed assumes the first three valid orders receive database IDs 1, 2 and 3. A clean reset is therefore recommended before rerunning the ETL.

The exact table name is orderitems. Order items are loaded after orders because they reference order IDs and product IDs. This sequencing preserves referential integrity.

5. Day 2 — SQL, Pandas and NumPy

Day 2 creates the analytical DataFrame directly from PostgreSQL through a three-table JOIN between orders, orderitems and products. The order date is converted with pandas and line_total is calculated as quantity × unit_price. The final five analytical rows are:

- Order 1 — 2024-09-01 — Completed — The Silent Orchard — Fiction — Qty 2 — 12.99 — Line total 25.98
- Order 1 — 2024-09-01 — Completed — Mystery at Midnight — Mystery — Qty 1 — 9.50 — Line total 9.50
- Order 2 — 2024-09-02 — Completed — Coding for Kids — Children's — Qty 1 — 15.00 — Line total 15.00
- Order 3 — 2024-09-03 — Pending — The Silent Orchard — Fiction — Qty 3 — 12.99 — Line total 38.97
- Order 3 — 2024-09-03 — Pending — Stars and Silence — Sci-Fi — Qty 2 — 11.25 — Line total 22.50

6. Day 2 Visual Analysis

Daily line values were 35.48 on 2024-09-01, 15.00 on 2024-09-02 and 61.47 on 2024-09-03. Cumulative values reached 111.95 by September 3. Category line value was: Fiction 64.95 (58.0%), Sci-Fi 22.50 (20.1%), Children's 15.00 (13.4%), and Mystery 9.50 (8.5%).

The five line totals range from 9.50 to 38.97, with an average of 22.39. The histogram communicates the distribution, while the scatter plot compares unit price and quantity. Because there are only five observations, the scatterplot demonstrates the method rather than supporting a strong statistical conclusion.

7. Day 3 — Seaborn, Waffle and Folium

The Day 2 category comparison is rebuilt with sns.barplot. Business sentence: Fiction generates the largest share of BookNook's analyzed line value.

A new Seaborn scatterplot uses unit_price on the x-axis, quantity on the y-axis and hue='category'. Business sentence: the plot shows how order behavior differs across categories, while the small sample limits the strength of any relationship claim.

The Waffle chart visualizes the revenue shares: Fiction 58.0%, Sci-Fi 20.1%, Children's 13.4% and Mystery 8.5%. Business sentence: Fiction accounts for more than half of the analyzed line value. The Folium map groups customers by city and scales CircleMarker size by customer count. Islamabad has 5 customers, Lahore 4,

Karachi 3, Peshawar 2, Multan 2, Rawalpindi 2, Faisalabad 1 and Quetta 1. The map is saved as booknook_customer_map.html.

8. Technical Lessons and Reproducibility

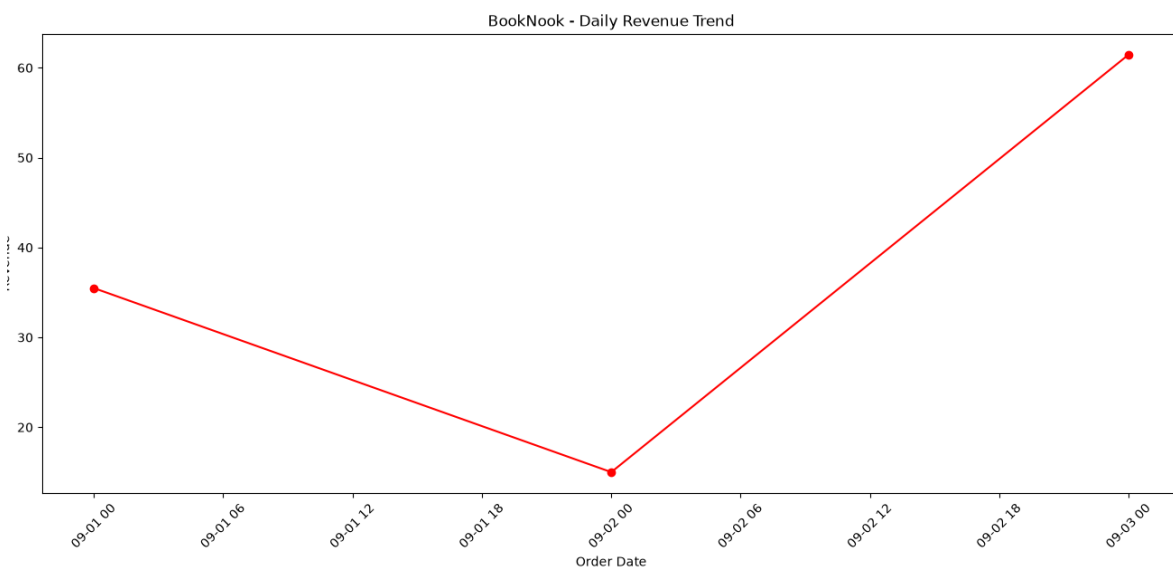
- Use orderitems exactly; do not substitute order_items.
- Load orders before orderitems.
- Reset/recreate tables for a clean rerun when fixed seed IDs are assumed.
- Convert order_date with `pd.to_datetime()` before time-based analysis.
- Make line_total explicit as $\text{quantity} \times \text{unit_price}$.
- Use `dropna()`, `loc/iloc` and `plt.show()` correctly.
- Folium coordinates must be [latitude, longitude].
- Keep PostgreSQL as the source of truth for Day 2/3 analysis.

9. Recommended Execution Order

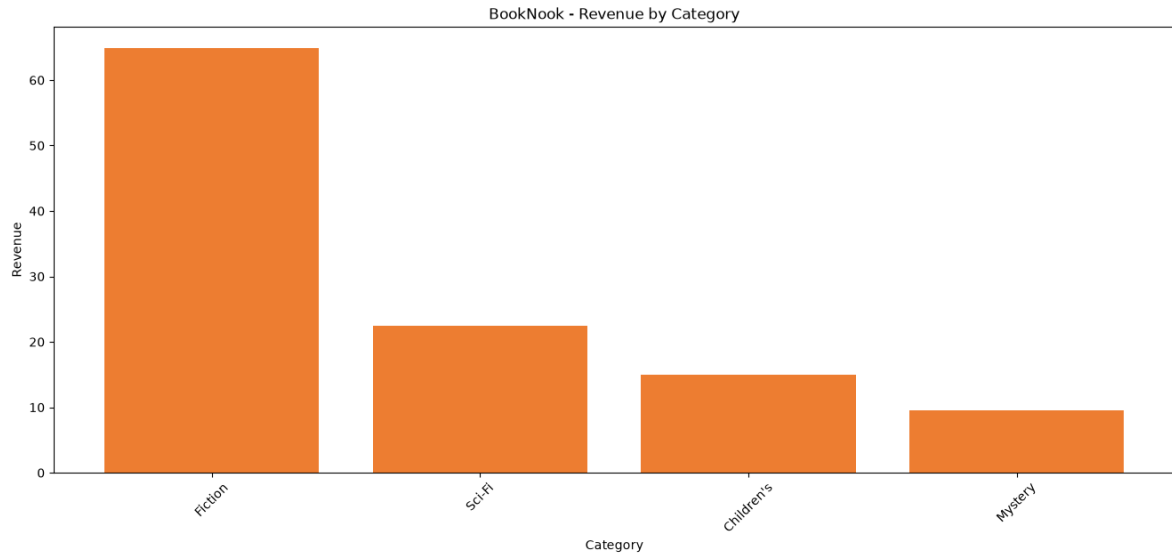
1. Create/reset the database tables.
2. Run customer/product ETL.
3. Run order transformation/loading.
4. Load orderitems after valid orders exist.
5. Run Day 2 analysis and visualization.
6. Run Day 3 Seaborn/Waffle/Folium visualization.
7. Verify booknook_customer_map.html.
8. Review every chart and add a plain-English business sentence.
9. Assemble and verify the final submission package.

10. Results:

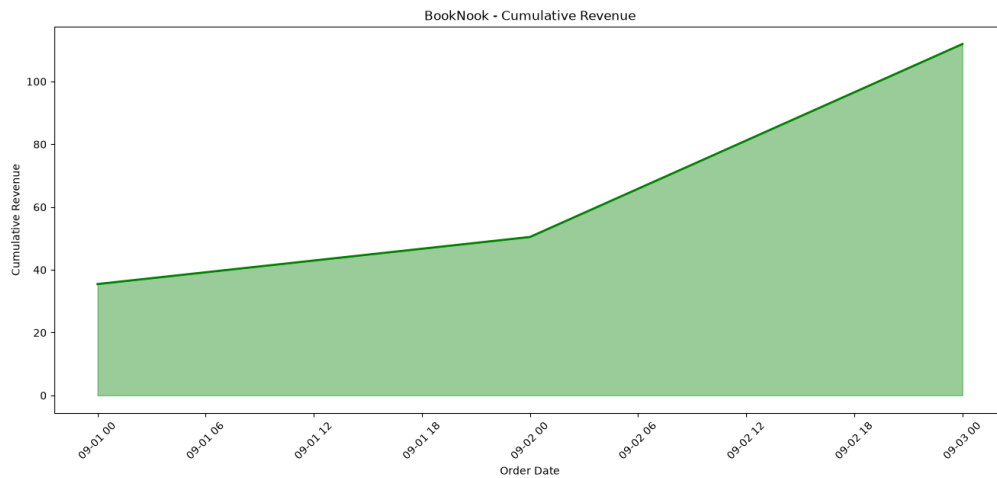
1. Daily Revenue Trend:



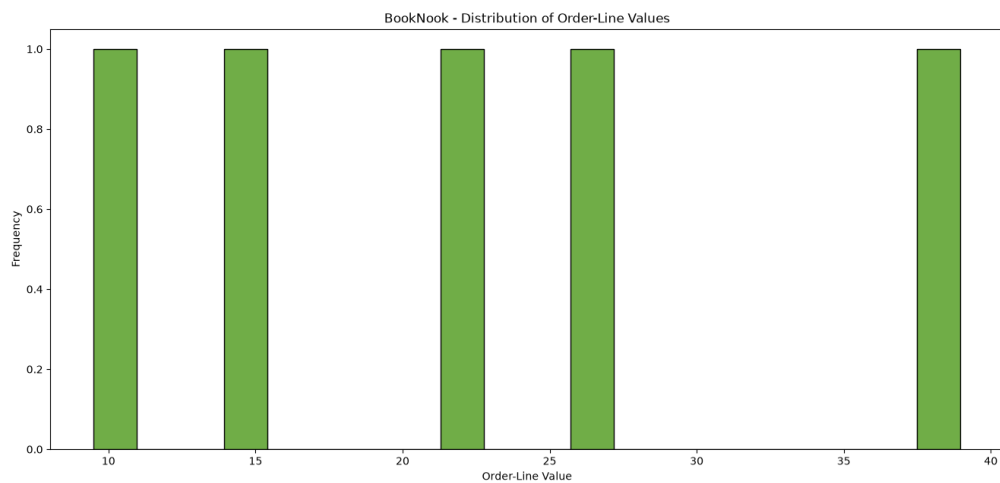
2. Revenue by Category



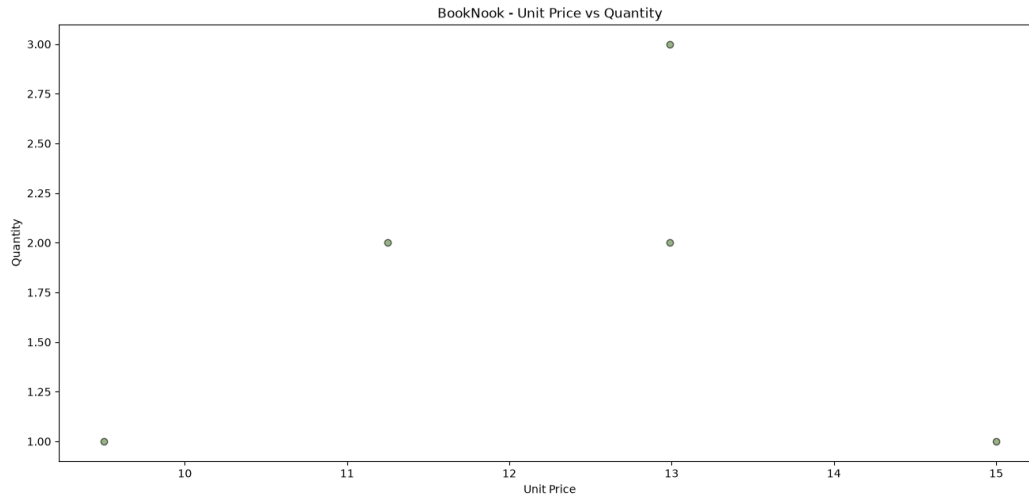
3. Cumulative Revenue



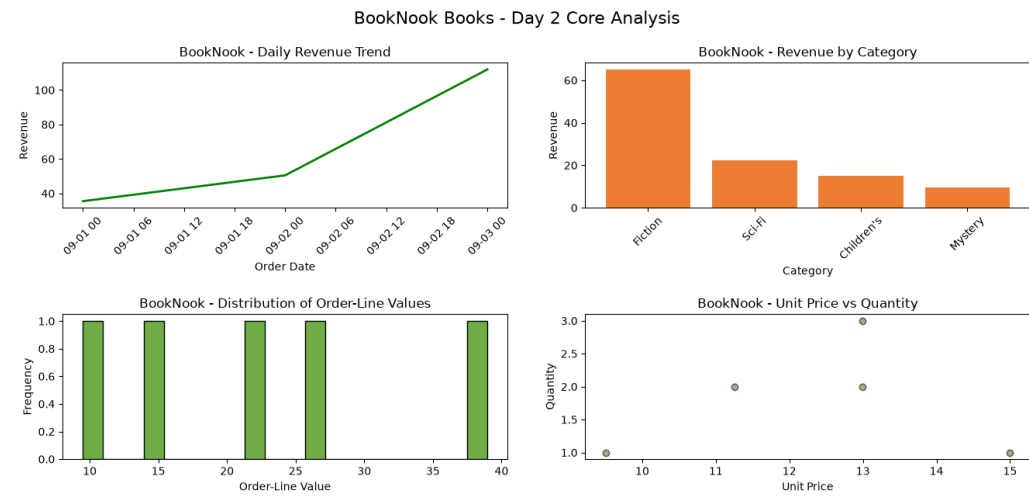
4. Distribution of Orderline Values



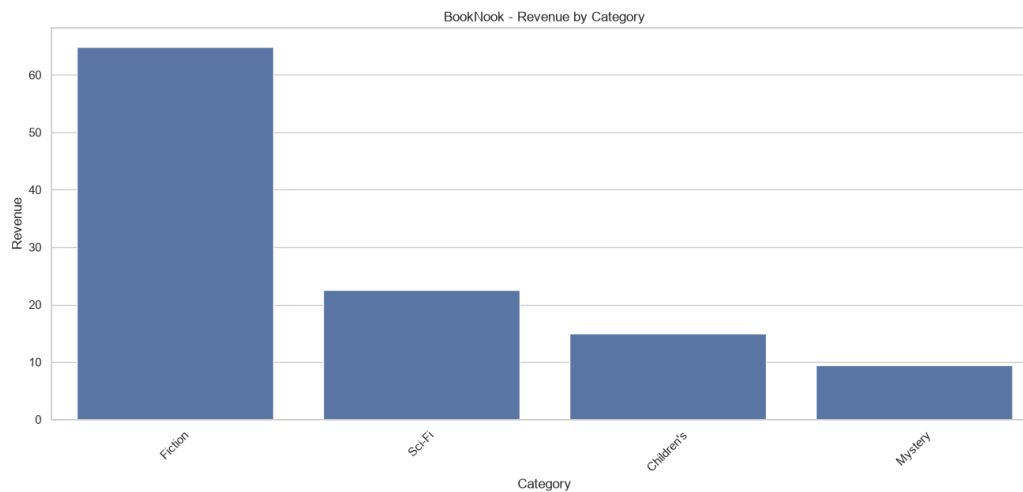
5. Scatter Plot: Unit Price vs Quantity



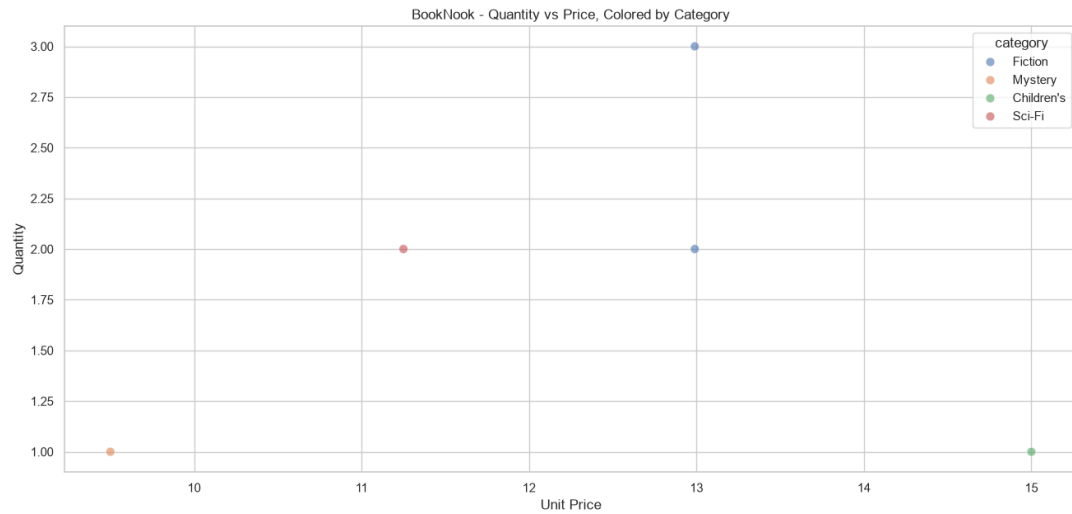
6. Subplots:



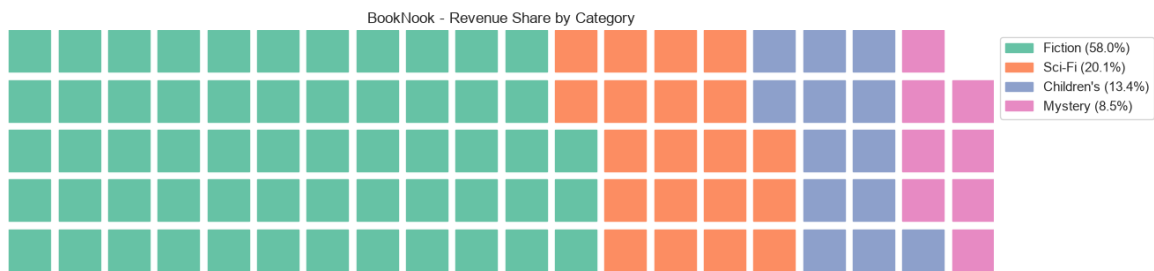
7. Bar Plot using Seaborn:



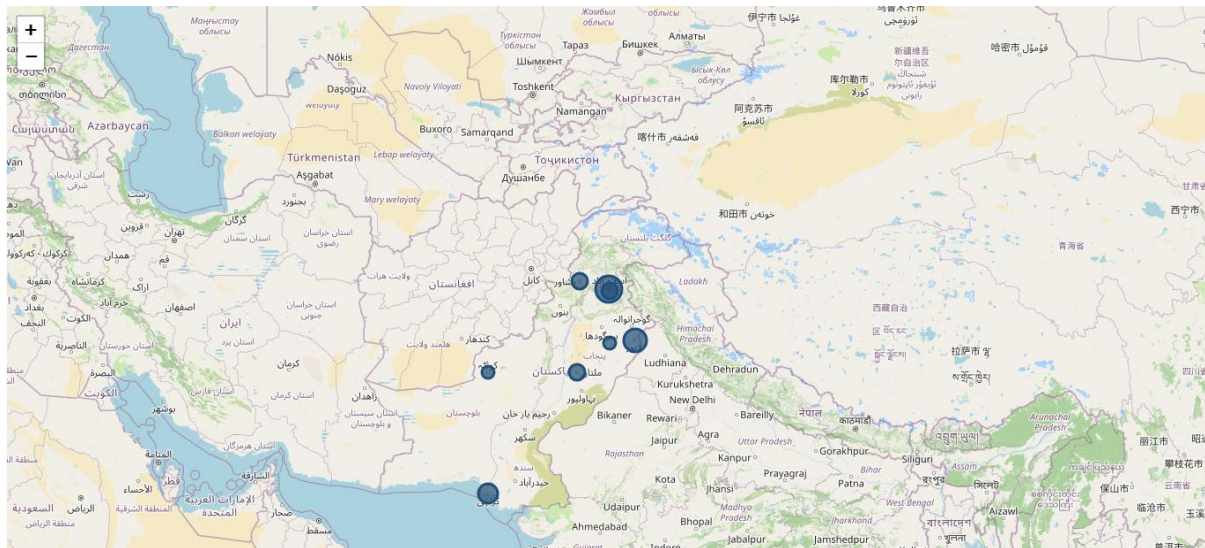
8. Scatter Plot using Seaborn:



9. Waffle Plot:



10. Folium Map:



13. Conclusion

BookNook connects database engineering with practical analytics in one reproducible workflow. It demonstrates how a small business dataset can be cleaned, stored, queried, analyzed and communicated using PostgreSQL and Python, ending with clear business-oriented visual storytelling.

Appendix A — NumPy Results

Metric	Result	Meaning
Total order lines	5	Five joined order-item records.
Total line value	111.95	Sum of quantity × unit price.
Mean order-line value	22.39	Average line value.
NumPy standard deviation	10.09	np.std() default ddof=0.
Lines above average	3	Three lines exceed 22.39.

Technical note: pandas describe() reports a larger standard deviation (about 11.28) because pandas uses ddof=1 for its sample standard deviation, while NumPy np.std() defaults to ddof=0. The project's requested NumPy result is 10.09.

Appendix B — Customer-City Counts

City	Customers
Islamabad	5
Lahore	4
Karachi	3
Peshawar	2
Multan	2
Rawalpindi	2
Faisalabad	1
Quetta	1